MLO 40" Observers Check Sheet Utilizing CCD2005

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Abstract

This check sheet is intended to aid in the operation of the Mount Laguna Observatory (MLO) 40" telescope utilizing the CCD2005 imaging camera. This check sheet is not a full operations manual and represents just one possible sequence of steps to successfully operate the telescope. It is assumed the observer has a working knowledge of observing and in particular the operation details of the 40". A more detailed 40" Observing Manual is available which provides detailed operating instructions and notes to assist in diagnosis. Section numbers (§) to the 40" Observing Manual are provided for reference. A simplified 40" Quick Start Sheet is also available for the more experienced observer.

This MLO 40" Check Sheet, the MLO 40" Observing Manual, the MLO 40" Quick Start Sheet, and blank Observing Log Sheets are available for download at http://mintaka.sdsu.edu/mlo/?.

1 Open The Observatory

- ☐ Be sure to lock the bottom floor entry door behind you after entering.
- ☐ Turn on the 4 light switches aside the entry door on the ground floor.

2 Set Up The Telescope

- ☐ Check the camera dewar has LN_2. *There should be a slight amount of fog coming out of the dewar. Check the observing log on the west wall as to when the dewar was last filled. The dewar has a 52 hour hold time.*
- ☐ Fill dewar if required. *Caution- finger tighten stinger, do not leave observatory floor while filling, aim exhaust over plywood. Fill out log on the west wall.*
- ☐ Plug in the power cord to the guider mirror power supply.
- ☐ Assure the observatory floor is clear. *The ladder is in the appropriate location and the filling dewar is moved out of the way.*
- ☐ Assure cabling is not tangled and is properly aligned on the cable managers.

3 Data Room Setup

- ☐ Check the observing log book for notes from prior observers. *In particular assess prior problems and resolutions.*
- ☐ Reboot the servers if necessary. *First the Telescope Control System (TCS) on the left, then the Instrument Server (IS) on the right.*

4 Set Up The Telescope Control System

- ☐ Log onto the Telescope Control System **[TCS]**, station on left.
- ☐ Start up ACE application *(If not running), dbl click the* **ACE release-date** *icon on the desktop.*
- ☐ Log onto ACE
- ☐ Open the focus menu **[TCS] / ACE / TELESCOPE / FOCUS.**
- ☐ Open the instrument menu **[TCS] / ACE / INSTRUMENTS / FILTER WHEEL.**
- ☐ Open the weather station **[TCS] / ACE / OBSERVATORY / WEATHER STATION.**

Note for interpreting this check sheet: **BOLD** text denotes an action or location such as a choose a radio button or look in a window. The forward slash "/" separates choices. Items underlined are to by typed in. Items both underlined and in *italic* are to be typed in, however the actual values will differ. [TCS] denotes the Telescope Control server and [IS] denotes the Instrument Server.

- ☐ Reset the guide camera pick off mirror position **[TCS] / ACE / AUXILIARY INSTRUMENTS / X-Y STAGE / POSITION / INITIALIZE.** *CAUTION- Do not initialize the focus stage.*
- ☐ Move the guide camera pick off mirror to a safe position **[TCS] / ACE / AUXILIARY INSTRUMENTS / X-Y STAGE / POSITION /** **45,1224** (green boxes) **/ GO-TO.**

5 Set Up The Instrument Server

- ☐ Log onto the Instrument Server **[IS]**, station on right.
- ☐ Start the OWL application *(If not running), dbl click the* **OWLversion** *(icon on the desktop).*
- ☐ Turn on and reset the CCD **[IS] / OWL / CONTROLLER / SETUP / APPLY / CLOSE.**
- ☐ Raise the camera temperature **[IS] / OWL / SUPPORTED CONFIGURATION / SIDIODE TEMPERATURE / SET ARRAY TEMP / -125 / RUN / CLOSE.**
- ☐ Monitor the camera temperature **[IS] / OWL / QUICK ACTIONS / THERMOMETER** *(icon with scroll)* **/ RUN** *(in plot window).*
- ☐ Check the readout binning **[IS] / OWL / SUPPORTED CONFIGURATION / BINNING SUPPORTED /** ***change if desired*** **/ RUN / CLOSE.**
- ☐ Check the sub-array **[IS] / OWL /SUPPORTED CONFIGURATION / SUB-ARRAY SUPPORTED /*****change if desired*** **/ RUN / CLOSE.**
- ☐ Start XWINSERVER **[IS] / START / PROGRAMS / CYGWIN-X / XWINSERVER.**
- ☐ Open an XTERM terminal **[IS] / START / PROGRAMS / CYGWIN-X / XTERM.**
- ☐ Move to the user data directory **[IS] / XTERM / % cd /E/images/*user_name***.
- ☐ Make a data directory **[IS] / XTERM / *user_name*% mkdir *yyyymmdd***.
  Note: yyyymmdd is UT for morning.
- ☐ Set the file directory to store images **[IS] / OWL / IMAGE FILE OPTIONS / DIR: E:\IMAGES*user_name**yyyymmdd*.** *Note: backslash "\" not "/".*
- ☐ Create the file naming structure **[IS] / OWL / IMAGE FILE OPTIONS / FILE: / *a1.fit***.
- ☐ Save and increment the file name **[IS] / OWL / IMAGE FILE OPTIONS / ✓SAVE** & **✓INCREMENT FILENAME.**
- ☐ Setup auto display **[IS] / OWL / IMAGE ANALYSIS / ✓DISPLAY** & **DS9 *icon*** *(choose yellow/green circle).*
- ☐ Enter observer names in header **[IS] / OWL / IMAGE FILE OPTIONS / click header icon / FITS HEADER / OBSERVER / *observer_name* / CLOSE.**
  Note: header icon is to the right of Increment Filename.
- ☐ Open an XGTERM session **[IS] / XTERM / % xgterm -sb &.**
- ☐ Move to the IRAF directory **[IS] / XGTERM / % cd /home/observer/iraf**.
- ☐ Launch IRAF **[IS] / XGTERM % cl.**
- ☐ Move to the data directory **[IS] /XGTERM > cd /E/images/*user_name/yyyymmdd*.**
- ☐ Launch ds9 **[IS] / XGTERM > !ds9 &.**
- ☐ Set the ds9 orientation **[IS] / DS9 / ZOOM /180.**
- ☐ Launch webcam to monitor cabling **[IS] / ???**.
  40-vid.mlo.sdsu.edu/eng/liveView.cgi

6 Set Up Your Laptop

- ☐ Launch JSkyCalc, load target objects, and start auto update.
- ☐ Monitor satellite radar- such as Hodar for Palomar
  http://observatories.hodar.com/palomar/index.html.
- ☐ Monitor seeing at Palomar
  http://www.palomar.caltech.edu:8000/maintenance/index.tcl.
- ☐ Monitor weather forecast- such as Acuweather
  http://www.accuweather.com/en/us/mount-laguna-ca/91948/hourly-weather-forecast/2178358.

7 Set Up Observing Log

- ☐ Sample observing logs can be downloaded from: http://mintaka.sdsu.edu/mlo/?
- ☐ Complete the observing log header information.

- ☐ Take the CCD temp [**IS**] / **OWL / QUICK ACTIONS / THERMOMETER** ***(icon).***
  Read temperature in [**IS**] / **OWL / CONTROLLER/ TEMPERATURE**.
- ☐ Record the temperature in observing **LOG / NOTES / CCD TEMP:** ***xxx*.**
- ☐ Turn off the temperature recording plot [**IS**] / **ARRAY TEMPERATURE / CLOSE.**
  Temperature plot used to contaminate image headers.

8 Take Bias

Confirm with PI how many biases to take, a typically recipe is 11 after the chip warms up to -125F and dwells for 20 minutes (early afternoon), 7 mid-run, and 11 at the end of the run.

- ☐ Best practice is to have dome lights off, dome shutter closed, and mirror cover closed.
- ☐ Set Name to bias [**TCS**] / **ACE / RA / NAME / BIAS.**
- ☐ Set the exposure time [**IS**] / **OWL / EXP TIME(S): / 0.**
- ☐ Keep the shutter closed [**IS**] / **OWL / EXPOSURE OPTIONS / UN-✓OPEN SHUTTER.**
- ☐ Set for single exposure [**IS**] / **OWL /EXPOSURE OPTIONS / UN-✓MULTIPLE EXPOSURE.**
- ☐ Start the exposure [**IS**] / **OWL / EXPOSE.**
- ☐ Inspect the image in ds9 [**IS**] / **DS9 / VALUES**
  Values should be $\sim$ *800 and image size should be 2200 X 2048.*
- ☐ Make the first entry in log sheet and mark as bad **LOG / CODE:** ***X*; FILE NAME:** ***a1*; OBJECT:** ***bias*; EXP TIME: 0; UT:** ***11:12*; PEAK:** ***800*; NOTES: CLEAR CHIP.**
  First image taken should be ignored as this is just to clear the chip.
- ☐ Set for multiple exposures [**IS**] / **OWL / EXPOSURE OPTIONS / ✓MULTIPLE EXPOSURE:** ***11*.**
- ☐ Start multiple exposures [**IS**] / **OWL / EXPOSE**.
- ☐ Make an entry in log sheet **LOG / CODE: B; FILE NAME:** ***a2-12*; OBJECT:** ***bias*; EXP TIME: 0; UTC:** ***11:12*; PEAK:** ***800*.**

9 Open The Observatory

- ☐ Open the dome shutters [**TCS**] / **ACE / DOME / OPEN SHUTTER / OPEN**.
  Wait, green status lights will say SHUTTER: OPEN and DROPOUT: OPEN when completed.
- ☐ Open the mirror cover [**TCS**] / **ACE / TELESCOPE / OPEN MIRROR COVER / YES.**
- ☐ Engage the auto-dome [**TCS**] / **ACE / DOME /AUTODOME / YES.**
- ☐ Engage the telescope tracking [**TCS**] / **ACE / TELESCOPE / TRACK ENABLE / YES.**
- ☐ Check all the status lights are green.
- ☐ Check the observing floor. *Telescope is pointing through the dome opening, cables are properly registered in the cable managers, and the floor is clear.*

10 Take Twilight Flats

The goal is to record at least 3 acceptable flats in each desired filter. It is possible to take 5 flats in 4 filters with reasonable counts if orchestrated efficiently. Begin with the shortest exposure on the first filter for 4 seconds. Move from shorter wavelength filters to longer wavelengths as the twilight darkens. Don't proceed much past an exposure time of 180 seconds. The desired counts are 15k-25k. Acceptable count range is 10k-55k. Be sure to offset between exposures such that star images can be removed with image processing.

- ☐ Point the telescope east of zenith [**TCS**] / **ACE / RA / RA:** ***sidereal + 1 hour*, DEC:** ***32 00 00*.**
- ☐ Move the telescope [**TCS**] / **ACE / GOTO / MOVE TELESCOPE.**
- ☐ Check the observing floor. *That the telescope is pointing east of zenith, pointing through the dome opening, the cables are properly registered, and lights are off.*
- ☐ Set the image name to twilight [**TCS**] / **ACE / RA / NAME /** ***TWILIGHT*.**
- ☐ Prepare for offset [**TCS**] / **ACE / OFFSET / RA /** ***50*.**
- ☐ Change the filter to shortest wavelength desired [**TCS**] / **ACE / AUXILIARY INSTRUMENTS / FILTER WHEEL /** ***B*.**
- ☐ Set the exposure time [**IS**] / **OWL / EXP. TIME(S):4**

- ☐ Reset back to single exposure [IS] / **OWL / EXPOSURE OPTIONS / UN-✓MULTIPLE EXPOSURE.**
- ☐ Set back to open shutter [IS] / **OWL / EXPOSURE OPTIONS / ✓OPEN SHUTTER.**
- ☐ Start the exposure [IS] / **OWL / EXPOSE.**
- ☐ Begin an entry in **LOG / FILE NAME:** ***a21***; **OBJECT: TWILIGHT; EXP. TIME:** ***4***; **FILTER:** ***B***; **UT:** ***10:20***.
- ☐ Offset the telescope [TCS] / **ACE / OFFSET / GOTO / OFFSET TELESCOPE.**
  Never offset during exposure. It is best practice to offset the telescope during the readout process.
- ☐ Read values in ds9 [IS] / **DS9 / VALUES**
  Look at vales of sky. Best practice is to keep at 20k.
- ☐ Record additional data in **LOG / PEAK:** ***15k***; **CODE:** ***F***; **NOTES:** ***3***.
  PEAK: sky values, Code F if an acceptable flat, X if unacceptable for counts or failure to offset; Notes: number of acceptable images for particular filter.
- ☐ Adjust the exposure time if necessary [IS] / **OWL / EXP TIME(S):** ***xx***.
- ☐ Change the filter after the desired number of acceptable images have been taken for the filter.
  Typically keep same exposure time when changing filters if previous exposure produced good counts. The exception is to reduce the exposure time by 20% when changing from R to V.
- ☐ Finish when all filters have been completed (minimum 3 acceptable exposures each filter) or exposure times are reaching too long > 180 seconds.

11 Focus Telescope *(Quimby/python method)*

- ☐ Copy focus.py into data directory. Example is from observer image directory above date directory from within IRAF.
  [IS] / **XGTERM** > **!cp ../focus.py .** **(note final period)**
- ☐ Open Cygwin terminal [IS] / **CYGWIN** *icon in system tray.*
- ☐ Change to data directory. [IS] / **CYGWIN / $** ***cd /E/images/chorst/20160301***
- ☐ Launch python [IS] / **CYGWIN / python**
- ☐ Load focus routine [IS] / **CYGWIN** / >>> / **import focus**
- ☐ Reset focus value, make large offset approximately 1000 below expected focus value. Ex assumes expected focus to be 14500. [TCS] / **ACE / FOCUS /** ***13500 (green box)*** **/ GOTO FOCUS**
- ☐ Bring focus back to approximately 300 below expected best focus value.
  [TCS] / **ACE / FOCUS /** ***14200 (green box)*** **/ GOTO FOCUS**
- ☐ Set focus jog to 50 or 100. [TCS] / **ACE / FOCUS /** ***50 (yellow box)***
- ☐ Locate field with a decent number of field stars such as an open cluster.
- ☐ Set sub array to interface with ds9. [IS] / **OWL / SUPPORTED CONFIGURATION / SUB ARRAY** ***icon*** **/ ✓GET COORDINATES FROM DS9 SELECTION BOX**
- ☐ Set to draw box on ds9. [IS] / **ds9 / REGION / SHAPE / BOX**
- ☐ Take a test image. [IS] / **OWL / EXP TIME / 10**
  Expose/ IS] / **OWL / EXPOSE**
- ☐ Draw box on ds9 about 700 by 700 pixels around a clean group of stars.
  [IS] / **ds9 /** ***click and drag to draw box***
- ☐ Apply sub array. [IS] / **SUBARRAY/ RUN**
  Note: this is currently not working. It may be necessary to enter in the sub array size in the X and Y of the red box. Then use dCol and dRow to position array. Note, the image is inverted.
- ☐ Check that sub array is sized and positioned properly by inspecting ds9 and making any necessary adjustments in the subarray and reexposing.
- ☐ Take a series of exposures and increase the focus value between exposures. For this example consider the focus evaluation to go from a23.fit to a32.fit.
  [IS] / **OWL / EXPOSE**
  [TCS] / **ACE / FOCUS / JOG+**
  Continue until approximately 300 points above expected good focus value.
- ☐ Compute focus. [IS] / **CYGWIN** / >>> **focus.check(*23*,32)**
  Read optimal focus value.

It is best to compute optimal focus from around 150-200 points above and the same amount above for the curve fitting to work well. It may be necessary to recompute and change the start and stop image numbers.

- ☐ Reset focus to optimal value. First make large step back to original far below value. Then go to newly derived focus value.
  [TCS] / ACE / FOCUS / *13500* *(green box)* / GOTO FOCUS
  [TCS] / ACE / FOCUS / *14322* *(green box)* / GOTO FOCUS *(For this example consider the optimal focus to be 14322*
- ☐ Reset sub array. **[IS] / SUBARRAY / RESET**

12 Locate Object

- ☐ Move the telescope to desired object **[TCS] / ACE / RA / NAME: *SN2015axz*; RA: *12:01:23.2*; DEC: *+32:00:00* / MOVE.**
  An alternate method is to use the catalog feature in ACE if the object exists in a predefined catalog. See the MLO Observing Manual for instructions in creating or editing a catalog. To open a catalog **[TCS] / ACE / *(rt-click in large black box)* / *(choose catalog)* / *(select object)* / GOTO.**
- ☐ Assure the telescope cables are aligned. *With each major move, either walk out onto the floor or utilize the webcam to inspect the cables. It is best practice to assure telescope is pointing through the opening in the dome.*
- ☐ Set short exposure time **[IS] / OWL / EXP TIME(S):10**
- ☐ Start the exposure **[IS] / OWL / EXPOSE**
- ☐ Make an initial log entry **LOG / FILE NAME, OBJECT, EXP TIME, FILTER, UT, AIRMASS.**
- ☐ Inspect the image after download. **[IS] / DS9.**
  Assure object is properly placed within the field of view and the counts are acceptable.
- ☐ *OPTIONAL* evaluate data in IRAF **[IS] / XGTERM /** > **imexam *a23*.**
  Check seeing/focus "FWHM" of stars/object and inspect for appropriate exposure time. Use imexam commands "a" and "r".
- ☐ Complete the log entry **LOG / CODE, FWHM, PEAK.**
- ☐ *OPTIONAL* offset telescope if necessary **[TCS] / ACE OFFSET / RA: *xx*; DEC: *xx* / MOVE.**
- ☐ *OPTIONAL* confirm move by acquiring and evaluating another short exposure.

13 Start Guider *(optional)*

- ☐ Startup the MAXIM DL application **[IS] / MAXIM DL** *icon on the desktop.*
- ☐ Start camera controller **[IS] / MAXIM / VIEW / CAMERA CONTROL.**
- ☐ Connect the camera **[IS] / MAXIM / CAMERA CONTROL / SETUP / CONNECT.**
- ☐ Turn the coolers on **[IS] / MAXIM / CAMERA CONTROL / SETUP / COOLERS ON.**
- ☐ Set for continuous exposure **[IS] / MAXIM / CAMERA CONTROL / EXPOSE / CONTINUOUS**, *seconds = 2.*
- ☐ Start continuous exposure **[IS] / MAXIM / CAMERA CONTROL / EXPOSE / START.**
- ☐ Turn the coolers on **[IS] / MAXIM / CAMERA CONTROL / COOLERS ON.**
- ☐ Move the pickoff mirror as required **[TCS] / ACE / AUXILIARY INST / X-Y STAGE / (N/S/E/W).**
  Set move length to 100 in yellow box in X-Y STAGE, adjust as required.
  Look for bright star in **[IS] / MAXIM / CCD IMAGE 2.**
- ☐ Stop continuous exposures **[IS] / MAXIM / CAMERA CONTROL / EXPOSE / STOP.**
- ☐ Set guider for single exposure **[IS] / MAXIM / CAMERA CONTROL / GUIDE / EXPOSE().**
- ☐ Take single guide exposure **[IS] / MAXIM / CAMERA CONTROL / GUIDE / START.**
- ☐ Open information window **[IS] / MAXIM / VIEW / INFORMATION WINDOW.**
- ☐ Choose guide star **[IS] / MAXIM / AUTOGUIDER IMAGE.**
  Dbl click periscope cursor on bright star.
- ☐ Set up to track **[IS] / MAXIM / CAMERA CONTROL / GUIDE / TRACK().**
- ☐ Start tracking **[IS] / MAXIM / CAMERA CONTROL / GUIDE / START.**

- ☐ Turn autoguiding on ACE [**TCS**] / **ACE / AUXILIARY INSTRUMENTS / X-Y STAGE / AUTOGUIDER / ON().**
  Check numbers are received in blue boxes in ACE.
- ☐ Start guiding error graph [**IS**] / **MAXIM / CAMERA CONTROL / GUIDE / GRAPH**.

14 Observe Objects

- ☐ Set the exposure time [**IS**] / **OWL / EXP. TIME(S): *180*.**
- ☐ Start the exposure [**IS**] / **OWL / EXPOSE.**
- ☐ Make an initial log entry **LOG / FILE NAME, OBJECT, FILTER, EXP TIME, UT, AIRMASS.**
- ☐ Evaluate the image after readout [**IS**] / **XGTERM** / > **imexam a24.**
  Use imexam to assess FWHM of object if stellar type if not check field stars, check PEAK of object.
- ☐ Complete the log entry **LOG / CODE, FWHM, PEAK, NOTES.**
- ☐ NOTE: Keep eye on sky. *Step outside periodically to assess weather conditions and seeing. Make notes in observing log.*
- ☐ NOTE: Keep eye on the weather sites.
- ☐ NOTE: Watch guiding and adjust aggressiveness if necessary. *See Section 13 above.*

15 Take Morning Twilight Flats *(optional)*

Only required if evening twilight flats were unsuccessful.

- ☐ The reverse of Section 10. Start with the I filter at 180 seconds. Progress to shorter times and shorter wavelengths.

16 Take Dome Flats *(optional)*

- ☐

17 Take Additional Biases *(*optional)

- ☐ Take additional biases if desired as prescribed in Section 8 above.

18 Download Data *(rsync Method)*

Data is stored on the instrument server E drive within the image directory by observer name then date. It is possible to copy the data via USB, sftp, and rsync. Instructions for rsync are provided below. Best practice is to run rsync periodically throughout the observing run.

- ☐ Make a directory on laptop to store data
- ☐ Lookup the IP address on laptop. *OSX / SYSTEM PREFERENCES / NETWORK / STATUS*
- ☐ Move to the data directory in terminal [**IS**] / **XTERM** / % **cd /E/images/observer_*name/date_ut*.**
- ☐ Run rsync in **XTERM**
  % **rsync -tv *.fit *chuckh@172.23.50.223:/Users/chuckh/MOLASUS/DATA/20150706***
  Note: IP addresses will change.

19 Shut Down Procedure

- ☐ Warm up the guider camera [**IS**] / **MAXIMDL / CAMERA CONTROL / SETUP / COOLERS / WARMUP.**
- ☐ Cool down the imaging camera [**IS**] / **OWL / SUPPORTED CONFIGURATION / SIDIODE TEMPERATURE / SET ARRAY TEMP / -145 / RUN / CLOSE.**
- ☐ Close the mirror cover [**TCS**] / **ACE / TELESCOPE / CLOSE MIRROR COVER.**
  It is necessary to wait for the mirror cover to close before closing the dome shutters.
- ☐ Log out of IRAF [**IS**] / **XGTERM** / > **logout.**
- ☐ Close the terminals on [**IS**] / **DS9, XTERM, XGTERM.**
- ☐ Close Cygwin terminal if opened for python method focusing. [**IS**] / **CYGWIN** / >>> **quit()**
- ☐ Disconnect the guider camera [**IS**] / **MAXIMDL / CAMERA CONTROL / SETUP / DISCONNECT.**

- ☐ Close down all the windows on [**IS**].
- ☐ Log off [**IS**].
- ☐ Move the dome to 180 [**TCS**] / **ACE /DOME / DOME AZIMUTH / <u>180</u> / MOVE.**
- ☐ Stow park the telescope [**TCS**] / **ACE / TELESCOPE / STOW PARK / MOVE.**
- ☐ Close the shutters [**TCS**] / **ACE / DOME / CLOSE SHUTTERS / CLOSE.**
  Assure mirror cover has completely closed prior to closing the shutters by checking status light TELESCOPE / COVERS: CLOSED.
- ☐ Check the observing floor that the telescope is parked east of the zenith, the dome is at 180 (or 0), and the dome shutters are closed.
- ☐ Unplug the power cord to the guider camera power supply at the telescope.
- ☐ Log off [**TCS**] / **ACE**.

20 Close Down The Observatory

- ☐ Fill out the observing log notebook. Observer names, general assessment of the night, seeing, and list any problems and their resolutions.
- ☐ Take out the trash and replace trash bags.
- ☐ Turn off the lights to the bathroom and data room.
- ☐ Assure the door to the catwalk is locked.
- ☐ Turn off the 4 light switches on the ground floor by the exit door.
- ☐ Lock the door behind you.